

Bus Reservation And Ticketing System

<https://github.com/aiyshwary/Bus-Reservation-and-Ticketing-System>

Java OOP Console Application

OVERVIEW

A Java-based console application that simulates a complete bus ticket reservation and billing workflow, supporting multi-passenger booking, seat management, and discount-based billing operations.

IMPACT

Built a fully functional console-based Bus Reservation and Ticketing System in Java with structured input validation, in-memory seat management, and automated billing with passenger-type discounts.

TECHNICAL WALKTHROUGH

"I built a console-based Bus Reservation and Ticketing System in Java that allows seat management, passenger booking, billing with discounts, and transaction tracking using

i) structured input validation and in-memory data

ii) storage. "

Problem statement

Manual ticket booking is error-prone and slow.

This project simulates a real-world bus reservation system where users can manage destinations, seat availability, passenger bookings, billing, and payment status through a menu-driven interface.

High-level system flow

Step 1: Authentication

- Simple login using username and password

- Prevents unauthorized access

"This simulates role-based access at a basic level."

Step 2: Main Menu

The system has 5 main modules

i) Destination Management

ii) Passenger Booking

iii) Billing & Payment

iv) View Passenger Details

v) Exit

Module-wise explanation

Destination Module

i) Displays available destinations

Shows

i) Fare per destination

ii) Remaining seat count

iii) Seats are initialized dynamically

“Seat availability updates in real time after each booking.”

Passenger Booking Module

This is the core logic.

What happens here

i) Passenger name validation

ii) Destination selection with validation

iii) Seat availability check

iv) Passenger count input

v) Discount handling

vi) Fare calculation

Business logic

Total Fare = (Regular passengers × fare) + (Discounted passengers × (fare × 0.80))

“This ensures accurate billing while enforcing seat limits.”

Billing & Payment Module

i) Search passenger by name

ii) Prevent double payment

iii) Accept payment

iv) Compute change

v) Mark passenger as PAID

“Once paid, the passenger status is locked to avoid duplicate billing.”

View Module

i) Search passenger details

Displays

i) Destination

ii) Passenger count

iii) Discount count

iv) Total fare

v) Payment status

vi) Amount paid and change

“This acts like a transaction history lookup.”

Exit Module

Gracefully exits the system

Confirms continuation

KEY DETAILS

1. Implements user authentication, seat selection, and multi-passenger booking in a single session.
2. Handles billing logic including discount tiers for different passenger categories.
3. Uses structured in-memory data storage for transaction tracking without a database dependency.
4. Validates all inputs to prevent double-booking and illegal seat selections.
5. Supports 5 Mindanao route destinations with preset fares and 20 seats each; applies a flat 20% discount for PWD, Student, and Senior Citizen passengers.
6. Offers a 5-option console menu: Destination (view routes and live seat counts), Passengers (book), Billing (payment with change), View (search and receipt), and Exit.
7. Enforces a 3-attempt login lockout; uses three parallel 2D arrays to store passenger name/destination/status, counts, and fare data.